

The quadratic formula

ANSWER KEY · FOR TEACHERS



Interactive version

REMEMBER

Coefficients and square roots

Level 1 · threshold

g9.qe.1

1.

Write $2x^2 = 7x + 4$ in the form $ax^2 + bx + c = 0$ with $a > 0$. Enter a , b and c , in this order, separated by semicolons.

Answer: $a = 2$, $b = -7$, $c = -4$

Level 1 · threshold

g8.sqrt

2.

Work out $\sqrt{9 + 16}$.

Answer: 5

WORKED EXAMPLES

Worked examples

YOUR TURN

Level 1 · threshold

g9.qe.6

Solve $x^2 + 3x - 4 = 0$ with the formula.

WORKED SOLUTION

$$D = 3^2 - 4 \cdot 1 \cdot (-4) = 25, \text{ so } x_1 = \frac{-3 - 5}{2} = -4 \text{ and } x_2 = \frac{-3 + 5}{2} = 1.$$

Answer: -4; 1

YOUR TURN

Level 2 · satisfactory

g9.qe.6

Solve $3x^2 - 5x - 2 = 0$ with the formula.

WORKED SOLUTION

$$D = (-5)^2 - 4 \cdot 3 \cdot (-2) = 49, \text{ so } x_1 = \frac{5 - 7}{6} = -\frac{1}{3} \text{ and } x_2 = \frac{5 + 7}{6} = 2.$$

Answer: $x_1 = -\frac{1}{3}$, $x_2 = 2$

CHECK

Check yourself

Level 1 · threshold g9.qe.6

1.

What is the discriminant of $3x^2 - 7x + 2 = 0$?

(A) -73

(B) 43

(C) 25 $- D = (-7)^2 -$
 $4 \cdot 3 \cdot 2 = 49 -$
 $24 = 25.$

(D) 73

Level 1 · threshold g9.qe.6

2.

Solve $2x^2 + x - 6 = 0$.

WORKED SOLUTION

$$D = 1^2 - 4 \cdot 2 \cdot (-6) = 49, \text{ so } x_1 = \frac{-1 - 7}{4} = -2 \text{ and } x_2 = \frac{-1 + 7}{4} = \frac{3}{2}.$$

Answer: $x_1 = -2$, $x_2 = \frac{3}{2}$

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3.

Tomas solved the same equation, $2x^2 + x - 6 = 0$, and got $x = 2$ and $x = -\frac{3}{2}$. Check his first root: substitute $x = 2$ into the left side $2x^2 + x - 6$. What number do you get?

Answer: 4

4.

Here are the lines of the proof, without the right-hand column. For each step (1)–(5), write what was done and why it is allowed.

$$ax^2 + bx + c = 0 \quad | \quad (1)$$

$$x^2 + \frac{b}{a}x + \frac{c}{a} = 0 \quad | \quad (2)$$

$$x^2 + \frac{b}{a}x = -\frac{c}{a} \quad | \quad (3)$$

$$x^2 + \frac{b}{a}x + \frac{b^2}{4a^2} = \frac{b^2}{4a^2} - \frac{c}{a} \quad | \quad (4)$$

$$\left(x + \frac{b}{2a}\right)^2 = \frac{b^2 - 4ac}{4a^2} \quad | \quad (5)$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

MARKING SCHEME: TICK WHAT YOUR SOLUTION HAS

- (1) Both sides are divided by a . This is allowed because $a \neq 0$. 1 pt
- (2) $\frac{c}{a}$ is subtracted from both sides; (3) $\left(\frac{b}{2a}\right)^2$ is added to both sides. The same number on both sides does not change the roots. 1 pt
- (4) The left side is written as $\left(x + \frac{b}{2a}\right)^2$ (lesson 1.2's rule, with $\frac{b}{a}$ in place of b), and the right side is brought to the denominator $4a^2$. 1 pt
- (5) The square root is taken, which is possible only when $D = b^2 - 4ac \geq 0$, because a square is never negative; the $\pm$ covers both signs. Then $\frac{b}{2a}$ is subtracted. 1 pt

WORKED SOLUTION

- Both sides are divided by a . Allowed, because $a \neq 0$ in a quadratic equation.
- $\frac{c}{a}$ is subtracted from both sides.
- The square of half the coefficient of x , $\left(\frac{b}{2a}\right)^2 = \frac{b^2}{4a^2}$, is added to both sides.
- The left side is a square, $\left(x + \frac{b}{2a}\right)^2$, by lesson 1.2's rule. The right side is written over the common denominator $4a^2$: $\frac{b^2 - 4ac}{4a^2}$.
- If $D = b^2 - 4ac \geq 0$, the square root of both sides is taken: $x + \frac{b}{2a} = \pm \frac{\sqrt{D}}{2a}$. (If $D < 0$, there are no real roots, because a square is never negative.) Then $\frac{b}{2a}$ is subtracted from both sides.

5.

Solve the equation $x^2 + 2x - 15 = 0$. Show how you find the discriminant, then find both roots.

MARKING SCHEME: TICK WHAT YOUR SOLUTION HAS

$$D = 2^2 - 4 \cdot 1 \cdot (-15) = 64. \quad 1 \text{ pt}$$

$$\text{One root: } \frac{-2 - 8}{2 \cdot 1} = -5. \text{ (Either root may be called } x_1 \text{.)} \quad 1 \text{ pt}$$

$$\text{The other root: } \frac{-2 + 8}{2 \cdot 1} = 3. \text{ If both roots have the same mistake, the two roots together get 1 point, not 2.} \quad 1 \text{ pt}$$

WORKED SOLUTION

$$D = 2^2 - 4 \cdot 1 \cdot (-15) = 4 + 60 = 64, \text{ and } \sqrt{64} = 8.$$

$$x_1 = \frac{-2 - 8}{2 \cdot 1} = -5, \quad x_2 = \frac{-2 + 8}{2 \cdot 1} = 3.$$

$$\text{Answer: } x_1 = -5, x_2 = 3.$$

6.

Prove the quadratic formula yourself. Start from $ax^2 + bx + c = 0$ with $a \neq 0$, make the four moves, and give a reason for every step.

MARKING SCHEME: TICK WHAT YOUR SOLUTION HAS

$$\text{Divides both sides by } a, \text{ with the reason } a \neq 0: x^2 + \frac{b}{a}x + \frac{c}{a} = 0. \quad 1 \text{ pt}$$

$$\text{Subtracts } \frac{c}{a} \text{ and adds } \frac{b^2}{4a^2} \text{ to both sides: } x^2 + \frac{b}{a}x + \frac{b^2}{4a^2} = \frac{b^2}{4a^2} - \frac{c}{a}. \quad 1 \text{ pt}$$

$$\text{Writes } \left(x + \frac{b}{2a}\right)^2 = \frac{b^2 - 4ac}{4a^2} = \frac{D}{4a^2}. \quad 1 \text{ pt}$$

$$\text{Takes the square root when } D \geq 0 \text{ (a square is never negative), and gets } x = \frac{-b \pm \sqrt{D}}{2a}. \quad 1 \text{ pt}$$

WORKED SOLUTION

$$1. a \neq 0, \text{ so divide both sides by } a: x^2 + \frac{b}{a}x + \frac{c}{a} = 0.$$

$$2. \text{ Subtract } \frac{c}{a} \text{ from both sides: } x^2 + \frac{b}{a}x = -\frac{c}{a}.$$

$$3. \text{ Half of } \frac{b}{a} \text{ is } \frac{b}{2a}. \text{ Add its square to both sides: } x^2 + \frac{b}{a}x + \frac{b^2}{4a^2} = \frac{b^2}{4a^2} - \frac{c}{a}.$$

$$4. \text{ The left side is a square, and the right side has the denominator } 4a^2: \left(x + \frac{b}{2a}\right)^2 = \frac{b^2 - 4ac}{4a^2} = \frac{D}{4a^2}.$$

$$5. \text{ If } D \geq 0, \text{ take the square root: } x + \frac{b}{2a} = \pm \frac{\sqrt{D}}{2a} \text{ (the } \pm \text{ covers both signs, also when } a < 0 \text{). If } D < 0, \text{ the right side is negative and there are no real roots.}$$

$$6. \text{ Subtract } \frac{b}{2a}: x = \frac{-b \pm \sqrt{D}}{2a}.$$

Two consecutive integers

Level 3 • basic

PUPP style • 4 points

g9.qe.6

The sum of the squares of two consecutive positive integers is 181.

- Call the smaller integer n . Write an equation for n .
- Show that the equation can be written as $n^2 + n - 90 = 0$.
- Find the two integers.

MARKING SCHEME: TICK WHAT YOUR SOLUTION HAS

- | | |
|--|------|
| a) $n^2 + (n + 1)^2 = 181$. | 1 pt |
| b) $2n^2 + 2n + 1 = 181$, so $2n^2 + 2n - 180 = 0$, and dividing by 2 gives $n^2 + n - 90 = 0$. | 1 pt |
| c) Solves the equation, with the formula ($D = 361$) or by factorising, $(n + 10)(n - 9) = 0$: $n = 9$ or $n = -10$. | 1 pt |
| c) Rejects $n = -10$ (it is not positive) and answers: 9 and 10. | 1 pt |

WORKED SOLUTION

- The integers are n and $n + 1$, so $n^2 + (n + 1)^2 = 181$.
- $n^2 + n^2 + 2n + 1 = 181$, so $2n^2 + 2n - 180 = 0$. Divide both sides by 2: $n^2 + n - 90 = 0$.
- $D = 1^2 - 4 \cdot 1 \cdot (-90) = 361$, and $\sqrt{361} = 19$. So $n_1 = \frac{-1 - 19}{2} = -10$ and $n_2 = \frac{-1 + 19}{2} = 9$.
The integers are positive, so $n = 9$. **Answer: 9 and 10.** Check: $81 + 100 = 181$ ✓.

TEACHER NOTES

For the teacher

Pupils can say: “I can explain where the quadratic formula comes from, step by step, and I can use it to solve an equation when the discriminant is a perfect square.”

Where this lesson sits. Lesson 1.2 made the four moves with numbers. Here the same moves are made once with letters. This is the proof the programme asks for (“*Irodoma ir taikoma...*”). Name the moves exactly as in lesson 1.2. All the discriminants in this lesson are perfect squares. Irrational roots are lesson 1.4; the cases $D = 0$ and $D < 0$ are lesson 1.5.

The step with $|a|$. $\sqrt{4a^2} = 2|a|$, and the $\pm$ makes the formula right for $a < 0$ as well. Most textbooks skip this step. Show it to strong pupils; for the others, “the $\pm$ covers both signs” is enough. A quick oral example: $-x^2 + 4x + 5 = 0$ (lesson 1.1’s (E) multiplied by -1) gives $x_{1,2} = \frac{-4 \pm 6}{-2}$, that is, 5 and -1 , the same roots as (E).

Labels. The formula sheet labels the roots $x_1 = \frac{-b - \sqrt{D}}{2a}$ and $x_2 = \frac{-b + \sqrt{D}}{2a}$, and this lesson follows it. The NŠA marking keys of 2025 and 2026 label them the other way round. Which root is called x_1 does not matter; both roots must be found.

Common mistakes and what helps

- $-b$ when b is negative: $b = -3$ gives $-b = 3$ (Example 2, t2, c2).
- Squaring a negative number, $(-7)^2 = 49$, and the sign of $-4ac$ (c1). Have pupils write every coefficient in brackets: $D = (-3)^2 - 4 \cdot 2 \cdot (-2)$.

- Dividing by 2 instead of $2a$, or dividing only $\sqrt{D}$ by $2a$ (t2, c2). Write the whole numerator over $2a$ first.
- Not checking. A substitution catches a wrong root at once (c3: Tomas's $x = 2$ gives 4, not 0).
- Reading a, b, c before the equation has the form $\dots = 0$ (r1; the garden's $x(x + 3) = 40$).
- Keeping a root that makes no sense in a word problem (Example 1, the exam task).

Key terms (EN • LT): discriminant • *diskriminantas*; quadratic formula • *kvadratinės lygties sprendinių formulė*; perfect square • *tikslusis kvadratas*; proof • *įrodymas*.

Timing (45 min):

- Remember: 3 min.
- Hook: 3 min.
- Explain: 12 min (the moves with numbers 2 min, with letters 4 min, the discriminant 2 min, the root and the formula 4 min). If time is short, show the $|a|$ remark only to strong pupils.
- Examples 1–2 with their “Your turn”: 11 min. If time is short, set t2 as homework.
- Check c1–c4 (the core: both outcomes; c4 asks for the reason behind each step of the proof): 10 min.
- Summary: 2 min.

The Check block has the core first (c1–c4, easy to hard), then two full-solution tasks: c5 is the PUPP 3-point pattern (“find D , then the roots”), and c6 asks for the whole proof, written without looking, with a reason for every step. Set them, and the exam task, as homework or for pupils who finish early.

Programme: 32.1.1.1 “*Įrodoma ir taikoma kvadratinės lygties sprendinių formulė.*” Every PUPP session has a 3-point item “find D , then the roots” (2025 main session, task 4: $x^2 + 3x - 4 = 0$; 2026 main session, task 4: $x^2 + 8x + 15 = 0$). It is marked 1 point for D and 1 point for each root, or 1 point for the roots if the same mistake is made in both. NŠA threshold minimum (2025) 2.1: solve a complete quadratic equation with integer coefficients, e.g. $2x^2 + 5x - 3 = 0$. Achievements A1, A3.